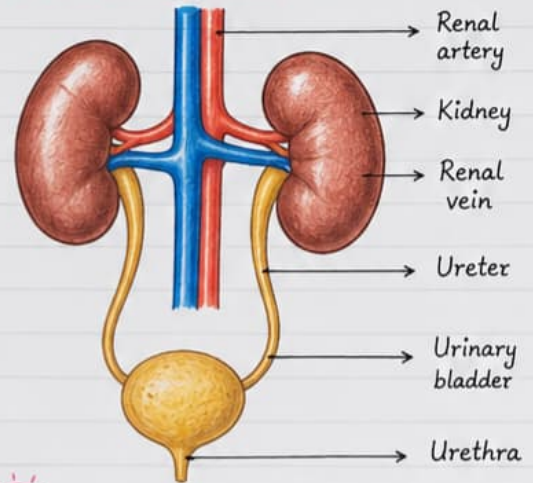


Excretion is the process of removal of metabolic waste products produced in the body. It is essential for maintaining **internal balance** and normal functioning of cells.

★ EXCRETORY PRODUCTS

1. **Carbon dioxide (CO_2)** – produced during respiration, eliminated through **lungs**.
2. **Water (H_2O)** – excess water is removed in **urine**, **sweat** and exhaled air.
3. **Nitrogenous wastes** –
 - **Ammonia (NH_3)** – in aquatic animals
 - **Urea** – in mammals
 - **Uric acid** – in birds, reptiles, insects
4. **Other wastes** – Bile pigments, excess salts, drug residues, hormones, etc.

★ HUMAN EXCRETORY SYSTEM

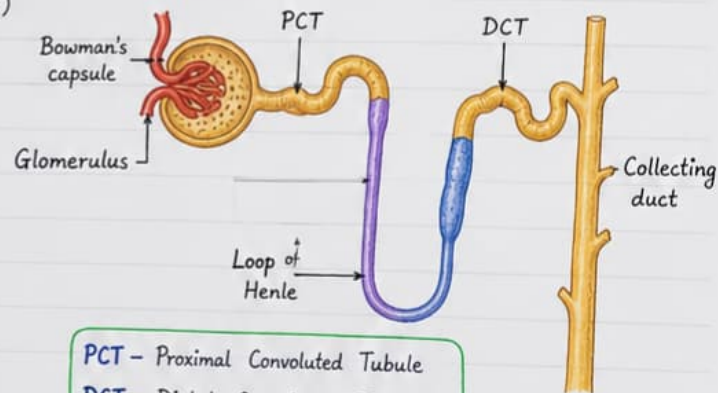


Kidneys are the main excretory organs in humans.

★ FUNCTIONS OF KIDNEYS

- Remove **nitrogenous wastes** (urea, uric acid, etc.)
- Regulate **water and electrolyte balance**
- Maintain **pH and osmotic balance**
- Help in **blood pressure regulation**
- Secrete hormones – **Erythropoietin, Renin**
- Activate Vitamin D

★ STRUCTURE OF NEPHRON



PCT – Proximal Convoluted Tubule
DCT – Distal Convoluted Tubule

★ STEPS OF URINE FORMATION

1. **Ultrafiltration** – Occurs in glomerulus. Water and small solutes filter into Bowman's capsule.
2. **Selective Reabsorption** – Useful substances (glucose, amino acids, ions, most of water) are reabsorbed back into blood.
3. **Tubular Secretion** – Certain wastes and excess ions are secreted from blood into tubule.

★ CONCENTRATION OF URINE

Juxtamedullary nephrons with long loop of Henle help in water reabsorption and concentrate the urine.

QUICK FACT

About 150-180 L of filtrate is formed daily, but only 1-1.5 L of urine is excreted.

Remember!

Excretion $\neq$ Egestion
Excretion – Metabolic waste
Egestion – Undigested food



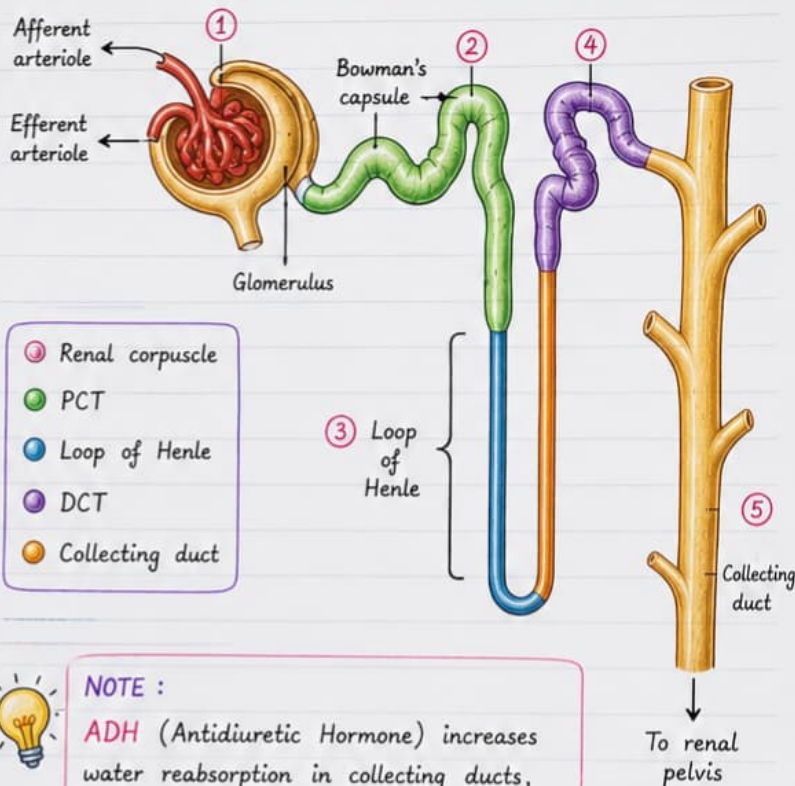
★ NEPHRON – STRUCTURAL AND FUNCTIONAL UNIT OF KIDNEY

Each kidney has about 1–1.5 million nephrons.

STRUCTURE OF A NEPHRON (L.S.)

PARTS OF NEPHRON

- ① **Renal corpuscle** – Consists of Glomerulus + Bowman's capsule. Site of **ultrafiltration**.
- ② **Proximal Convoluted Tubule (PCT)** – Reabsorption of glucose, amino acids, ions, water.
- ③ **Loop of Henle** –
 - Descending limb – permeable to water.
 - Ascending limb – impermeable to water, reabsorbs Na^+ , K^+ , Cl^- .
- ④ **Distal Convoluted Tubule (DCT)** – Selective reabsorption and secretion of ions (H^+ , K^+) and water.
- ⑤ **Collecting Duct** – Final reabsorption of water under the influence of **ADH**; urine is formed.



- ① Renal corpuscle
- ② PCT
- ③ Loop of Henle
- ④ DCT
- ⑤ Collecting duct



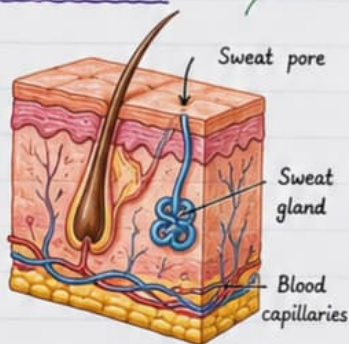
NOTE :

ADH (Antidiuretic Hormone) increases water reabsorption in collecting ducts, concentrating the urine.

ROLE OF OTHER ORGANS IN EXCRETION

1. SKIN (SWEAT GLANDS)

- Sweat contains water, salts, **urea**, lactic acid.
- Helps in excretion and regulation of body temperature.



2. LUNGS



- **Carbon dioxide (CO_2)** – major waste of cellular respiration.
- **Water vapour** and traces of volatile substances are also eliminated.

COMPOSITION OF URINE

- 95% **Water**
- 2.5% **Urea**
- 2.5% **Other solutes** – Uric acid, creatinine, excess salts (Na^+ , K^+ , Cl^- , PO_4^{3-} , SO_4^{2-}), pigments.



REGULATION OF EXCRETION

- The **hypothalamus** maintains water balance.
- **ADH** from posterior pituitary increases water reabsorption.
- **Aldosterone** (from adrenal cortex) increases Na^+ reabsorption and K^+ secretion.
- **ANP** (from heart) reduces Na^+ reabsorption, increases urine output.

★ SUMMARY

Excretion removes metabolic wastes like **urea**, CO_2 , **uric acid** and excess water and salts. **Kidneys** are the main excretory organs in humans. **Nephrons** filter blood, reabsorb useful substances and secrete wastes to form urine. **Skin** and **lungs** also help in eliminating certain wastes.

- | | |
|----------|---|
| Filter | ✓ |
| Reabsorb | ✓ |
| Secrete | ✓ |
| Excrete | ✓ |



Take care of your **kidneys**, they work **24x7** for you!



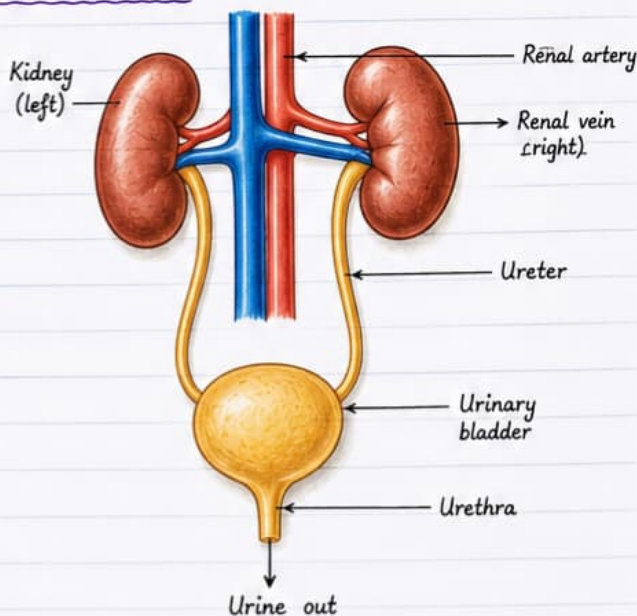
★ HUMAN EXCRETORY SYSTEM - DETAILED STUDY

COMPONENTS

- A pair of kidneys
- A pair of ureters
- Urinary bladder
- Urethra



The excretory system filters blood, removes wastes and maintains the internal balance of the body. ♥

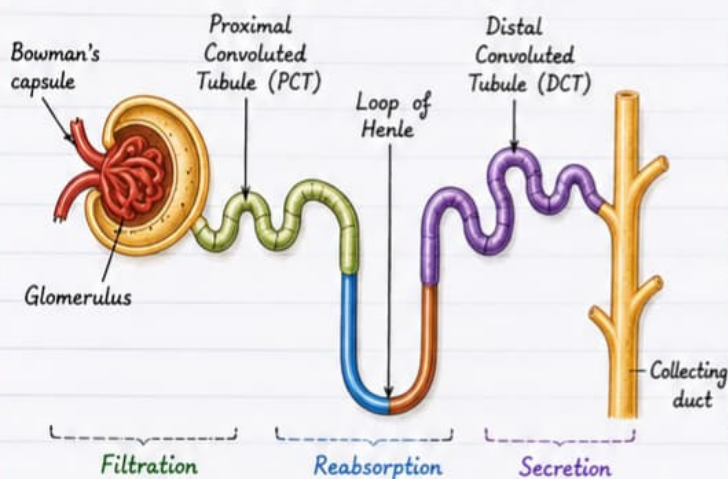


KIDNEYS - THE FILTERS

- Bean shaped organs, about **11-12 cm** long.
- Located on either side of vertebral column, below the rib cage.
- Receive impure blood from **renal artery**.
- Clean the blood and form urine.
- Each kidney has about **1-1.5 million** nephrons.



NEPHRON - STRUCTURAL AND FUNCTIONAL UNIT



URINE FORMATION - 3 MAIN PROCESSES

1. Glomerular Filtration (Ultrafiltration)

- Occurs in glomerulus.
- Water and small solutes (glucose, amino acids, salts, urea) pass into Bowman's capsule.

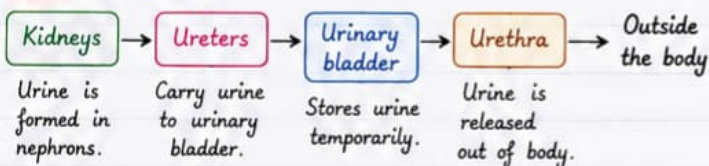
2. Selective Reabsorption

- Useful substances and water are reabsorbed back into blood from PCT, Loop of Henle and DCT.
- Helps in maintaining balance of fluids and electrolytes.

3. Tubular Secretion

- H^+ , K^+ , NH_3 , drugs and other ions are secreted from blood into tubule.
- Helps in removing additional wastes and regulating pH.

PATH OF URINE



OTHER ORGANS INVOLVED IN EXCRETION

- **Lungs** - Remove CO_2 and water vapour during exhalation. 
- **Skin** - Sweat glands excrete water, salts and small amounts of urea. 
- **Liver** - Converts toxic ammonia to urea and helps in detoxification. 
- **Large Intestine** - Eliminates undigested food and some water as faeces. 

IMPORTANCE OF EXCRETION

- ✓ Removes toxic wastes.
- ✓ Maintains water, salt and electrolyte balance.
- ✓ Maintains osmotic balance and pH.
- ✓ Helps in regulation of blood volume and pressure.
- ✓ Essential for overall health and survival. 



Excretion is not just waste removal, it is the **KEY** to internal balance and healthy life.

Remember!

- ★ Clean Blood → Healthy Body
- ★ Healthy Kidneys → Long Life



REGULATION OF EXCRETION

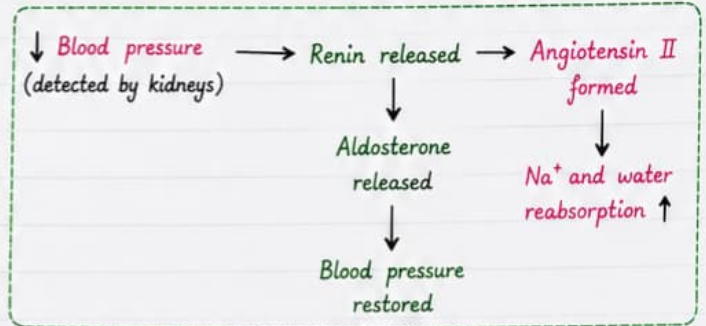
Excretion is a tightly regulated process controlled by the nervous and endocrine systems to maintain **homeostasis**.

HOMEOSTASIS

The tendency to maintain a stable internal environment despite changes in external conditions.

① NERVOUS REGULATION

- Sympathetic nervous system stimulates secretion of **renin** from kidneys.
- Helps in maintaining blood pressure and water balance.



② HORMONAL REGULATION

- **ADH** (Antidiuretic Hormone) – Increases water reabsorption in collecting ducts.
- **Aldosterone** – Increases reabsorption of Na^+ and secretion of K^+ .
- **ANP** (Atrial Natriuretic Peptide) – Released from heart, decreases Na^+ reabsorption and increases urine output.

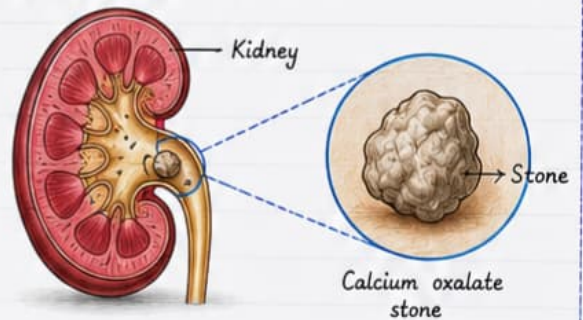
KEY HORMONES INVOLVED

Hormone	Source	Function
ADH	Posterior pituitary	↑ Water reabsorption, ↓ urine output
Aldosterone	Adrenal cortex	↑ Na^+ reabsorption, ↑ K^+ secretion
ANP	Heart atria	↓ Na^+ reabsorption, ↑ urine output

DISORDERS OF EXCRETORY SYSTEM

- **Renal Failure** – Kidneys lose the ability to filter wastes; dialysis or transplant needed.
- **Kidney Stones** – Hard deposits of salts formed in kidneys causing severe pain.
- **Urinary Tract Infection (UTI)** – Caused by bacteria; burning sensation during urination.
- **Gout** – Due to excess uric acid deposition in joints.

KIDNEY STONE (3D VIEW)



Key Takeaway Excretion is vital for removing harmful wastes and maintaining internal balance. The kidneys are the main excretory organs in humans, working with other systems for overall health.



QUICK RECAP

- **Wastes:** CO_2 , H_2O , Urea, Uric acid, etc.
- **Main Organ:** Kidneys
- **Urine Formation:** Filtration → Reabsorption → Secretion
- **Hormones:** ADH, Aldosterone, ANP
- **Functions:** Remove wastes, maintain balance of water, salts, pH and blood pressure.

MEMORY TIP

F.R.E.S.H for Excretion

- F – Filtration
- R – Reabsorption
- E – Excretion (Secretion)
- S – Secretion (Tubular)
- H – Homeostasis Maintained



“A healthy body is the result of proper excretion and a balanced life.”

